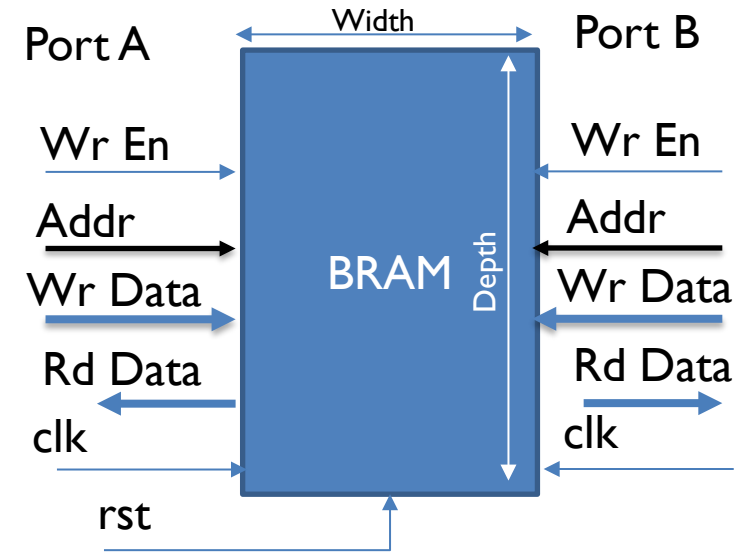
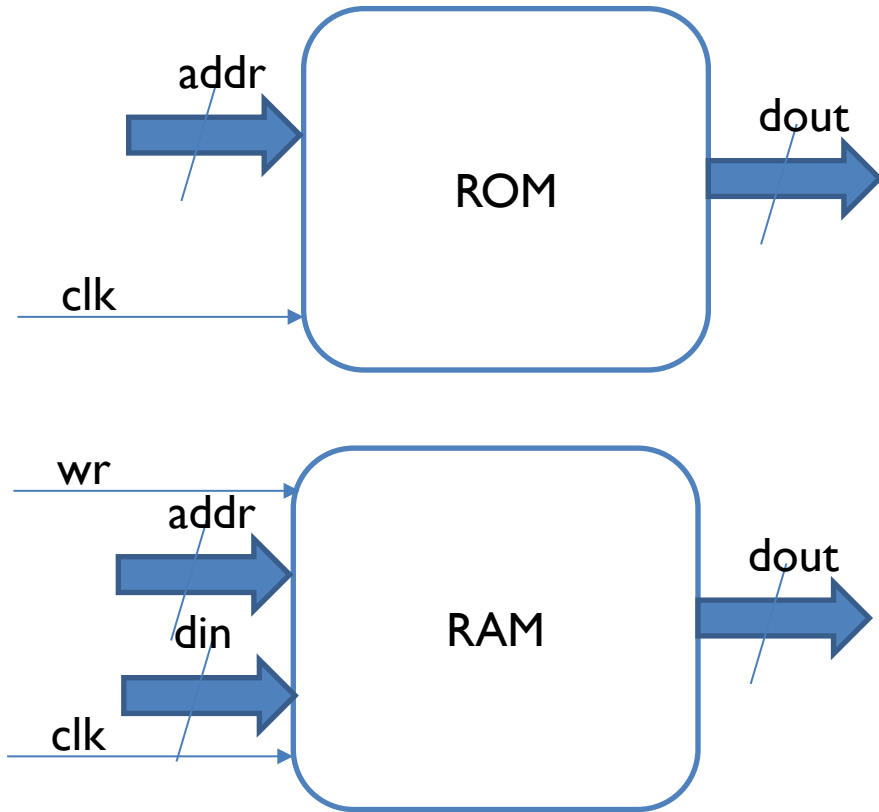


Lecture II:
Error detection in HDL &
Digital Design Performance: Metastability and Clock Skew

By: Emad Samuel Malki Ebeid

Summary of lecture 10

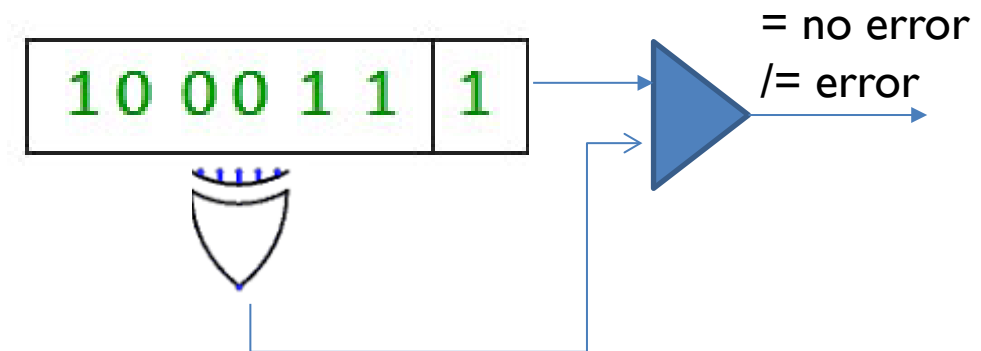
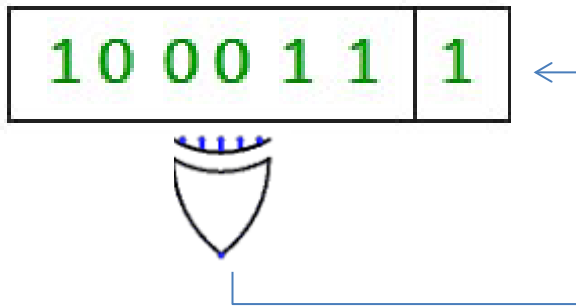


Error Detection

- During transmission, digital signals may be corrupted due to noise that introduces errors when the signal is traveling between the sender to the receiver (e.g., 0 becomes 1 and 1 becomes 0).
- Error detection code adds additional data to a given digital message to help detect if any error has occurred during the transmission.
- Popular techniques for error detection are:
 - Simple Parity check – as in UART
 - Checksum
 - Cyclic redundancy check

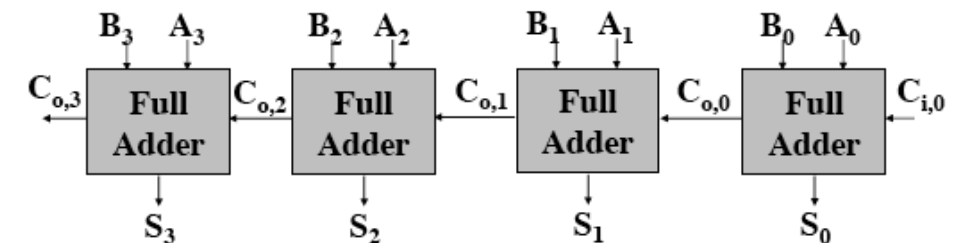
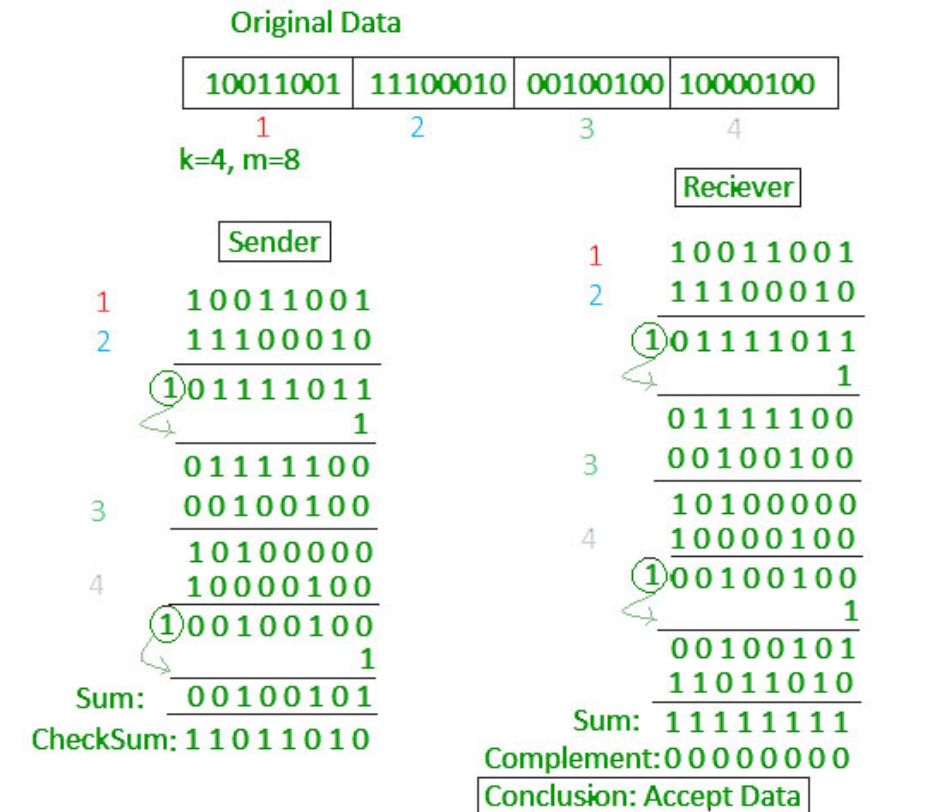
Simple Parity check

- It is a check bit or parity bit generated from the original data, where parity of :
 - 1 is added to the data if it contains an odd number of 1's, and
 - 0 is added if it contains an even number of 1's
- XOR gate is used to create the parity bit
- Example:



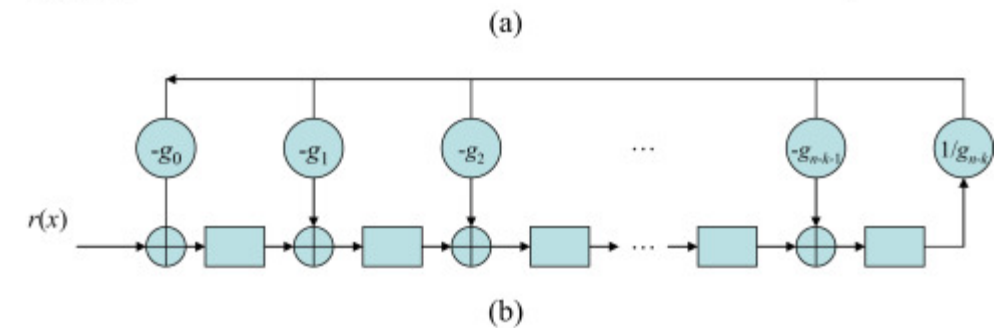
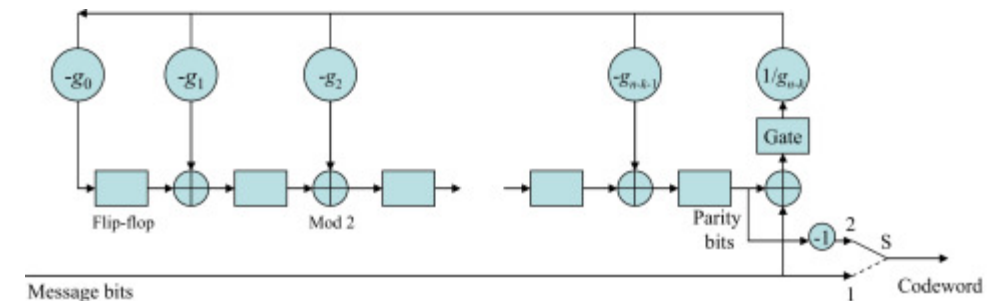
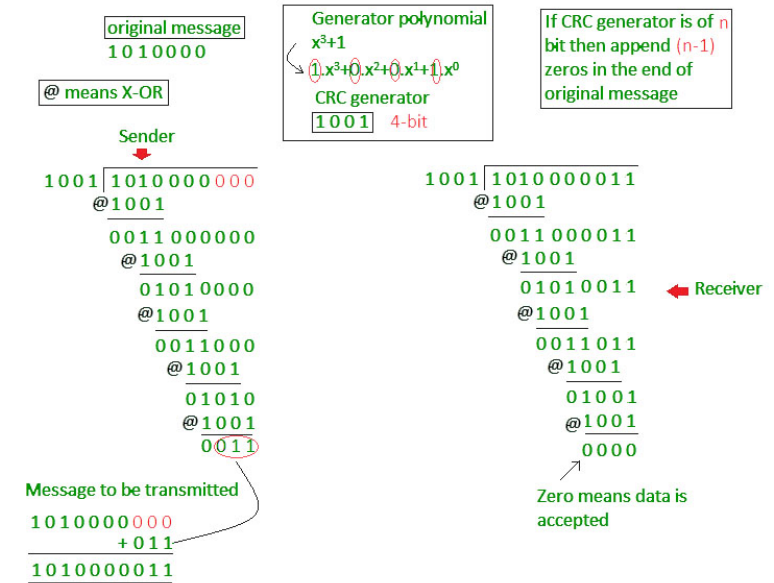
Checksum

- the data is divided into k segments each of m bits.
- In the sender's end the segments are added to get the sum.
 - The sum is complemented to get the checksum.
- At the receiver's end, all received segments are added to get the sum.
 - The sum is complemented.
 - If the result is zero, the received data is accepted; otherwise discarded.



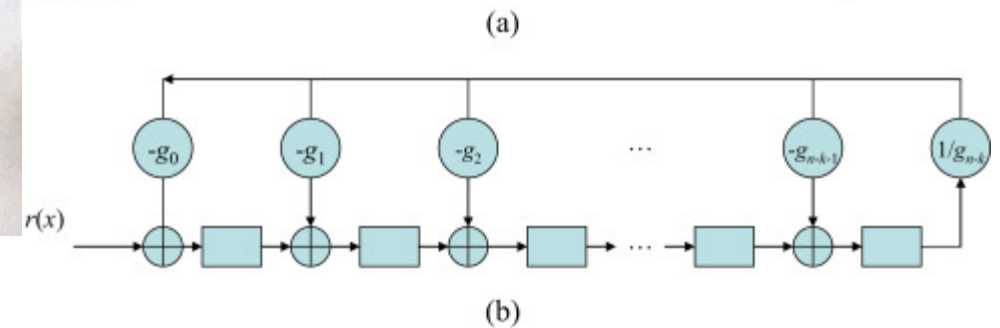
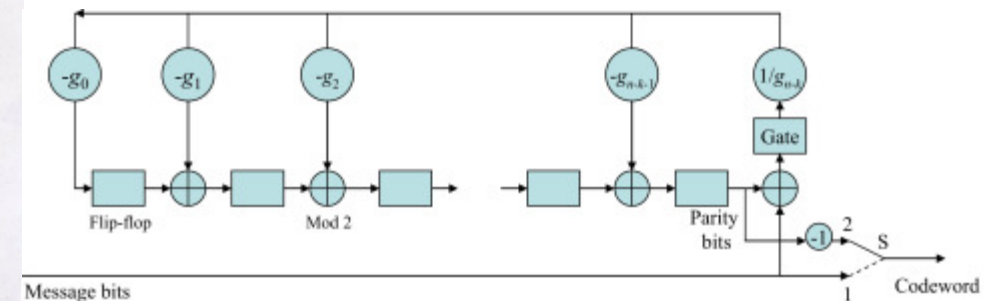
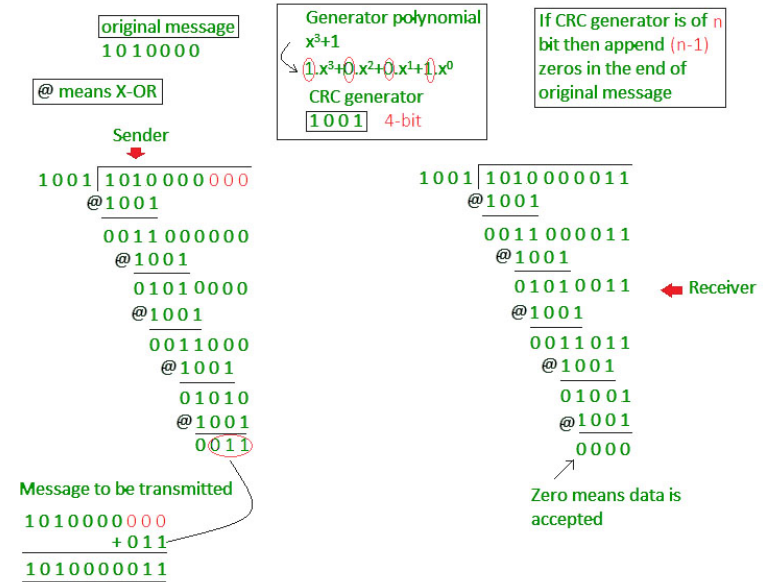
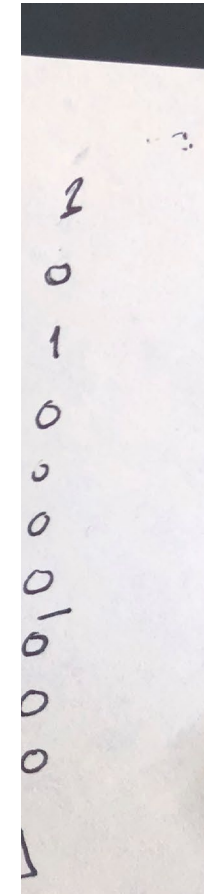
Cyclic redundancy check (CRC)

- It is based on binary division.
- The cyclic redundancy check bits, are added to the end of the data unit.
- The CRC bits are the remainder of the division in which the divisor is prechosen.
- At the receiver, the data is divided over the same divisor.
- Polynomial generators based on the divisor value are used to generate the CRC bits.
- In HDL, shift registers with XORs are used to build the CRC circuit



Cyclic redundancy check (CRC)

- It is based on binary division.
- The cyclic redundancy check bits, are added to the end of the data unit.
- The CRC bits are the remainder of the division in which the divisor is prechosen.
- At the receiver, the data is divided over the same divisor.
- Polynomial generators based on the divisor value are used to generate the CRC bits.



Digital Design Performance: Metastability and Clock Skew

Quality Metrics of a Digital Design

- Quality metrics help to quantify the quality of a design from different perspectives:
 - Performance, energy consumption, cost, functionality, and robustness.
- Which one of these metrics is most important?

It depends upon the application.

 - Server vs hand-held mobile.
 - Ground robot vs aerial robot (drone)

Performance

- The performance of a digital circuit expresses the computational load that the circuit can manage.
 - the duration of the clock period (*clock cycle time*) or clock rate (*clock frequency*)
- For instance, a microprocessor is often characterized by the number of instructions it can execute per second.
- Circuit performance depends on the architecture of the processor (the number of instructions it can execute in parallel) and the *actual design of logic circuitry*.

Propagation delay

- The *propagation delay* t_p of a gate defines how quickly it responds to a change at its input(s)
- $t_p = \frac{t_{pLH} + t_{pHL}}{2}$
- Oscillation period (T) =
 $2 * t_p * N$ (only valide if $T \gg t_f + t_r$)
 - Typically, a ring oscillator needs a least five stages to be operational
- Clock Frequency = $\frac{1}{T}$
 - If $T = 2ns \rightarrow F = 0.5GHz$

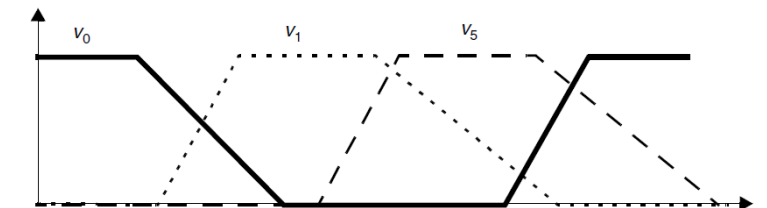
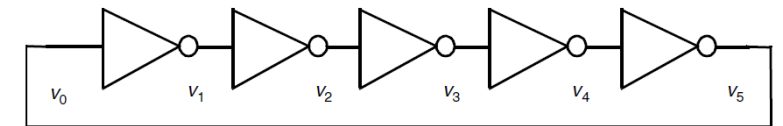
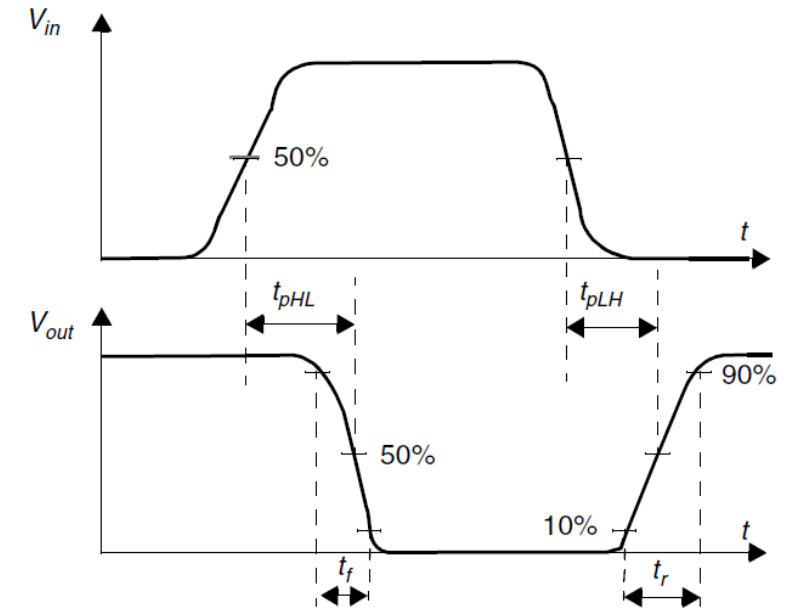
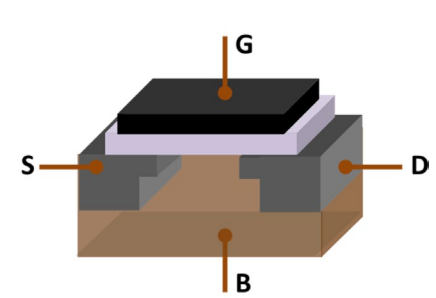
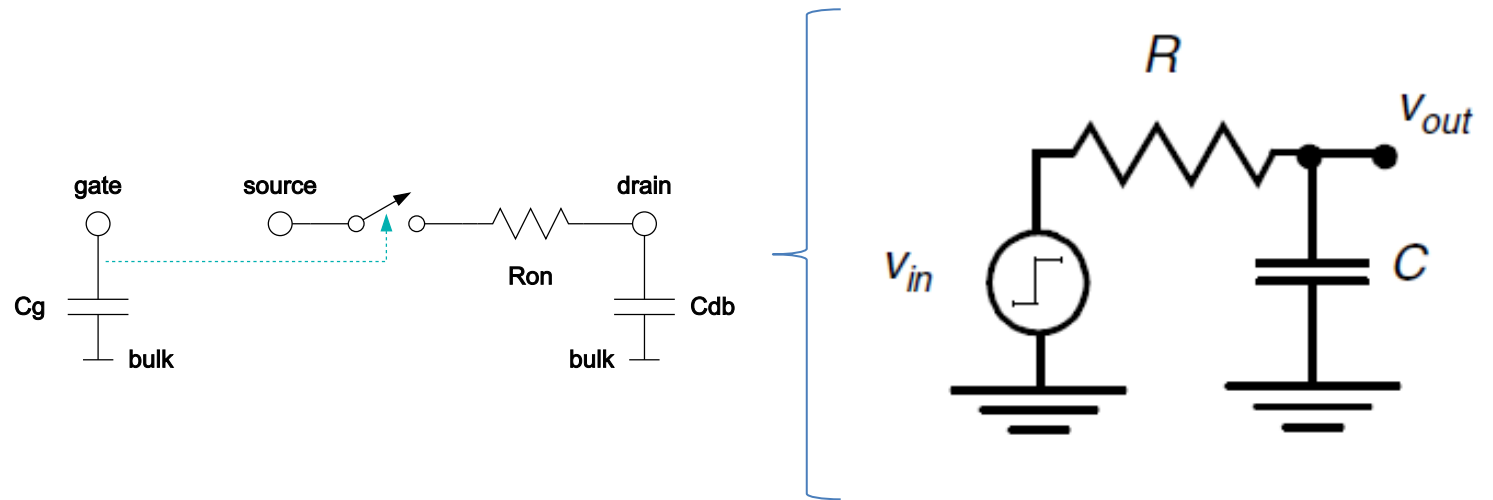
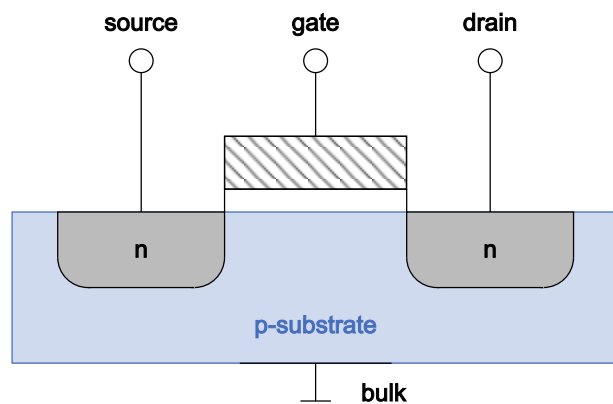


Figure 1.20 Ring oscillator circuit for propagation-delay measurement.

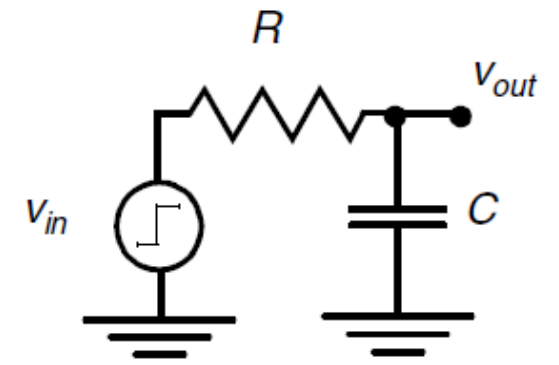


Propagation delay of the first-order RC networks

- The RC step response is a fundamental behavior of all digital circuits and thus digital circuits are often modeled as first-order RC networks.
- The propagation delay of the circuit is of considerable interest.



Propagation delay



First order RC network

$V_{in} = V_m \sin(\omega t)$

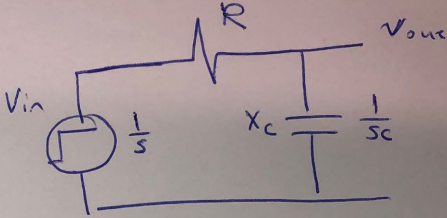
Group work: calculate the $V_{out}(t)$
(15 minutes)

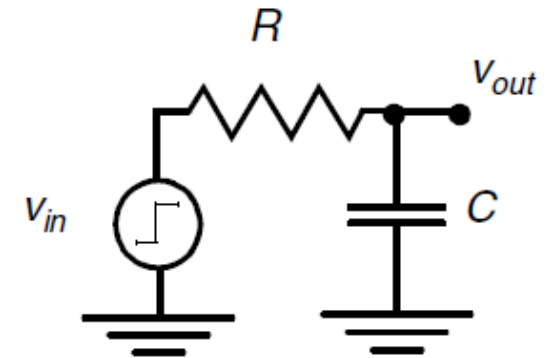
Propagation delay

First order RC network (another solution)

$$\begin{aligned}
 V_{out} &= V_{in} \frac{X_c}{R + X_c} \stackrel{s \text{ domain}}{=} \frac{1}{s} \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{s} \cdot \frac{1}{sRC + 1} \\
 &= \frac{A}{s} + \frac{B}{sRC + 1} \quad \Leftarrow \text{Partial fraction} \\
 A \Big|_{s=0} &= 1 \qquad B \Big|_{s=-\frac{1}{RC}} = -RC \\
 &= \frac{1}{s} + \frac{-RC}{sRC + 1} = \frac{1}{s} - \frac{1}{s + \frac{1}{RC}} \\
 V_{out} &= (1 - e^{-\frac{t}{RC}}) V_{in}
 \end{aligned}$$

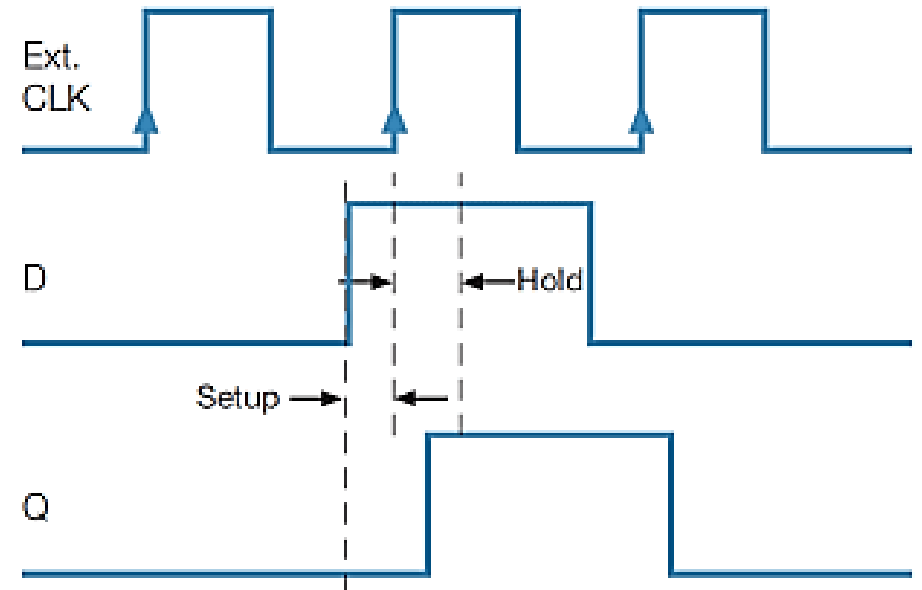
Laplace transform:-

$$\begin{aligned}
 V(s) &= \frac{1}{s} \\
 e^{-\alpha t} &= \frac{1}{s + \alpha}
 \end{aligned}$$




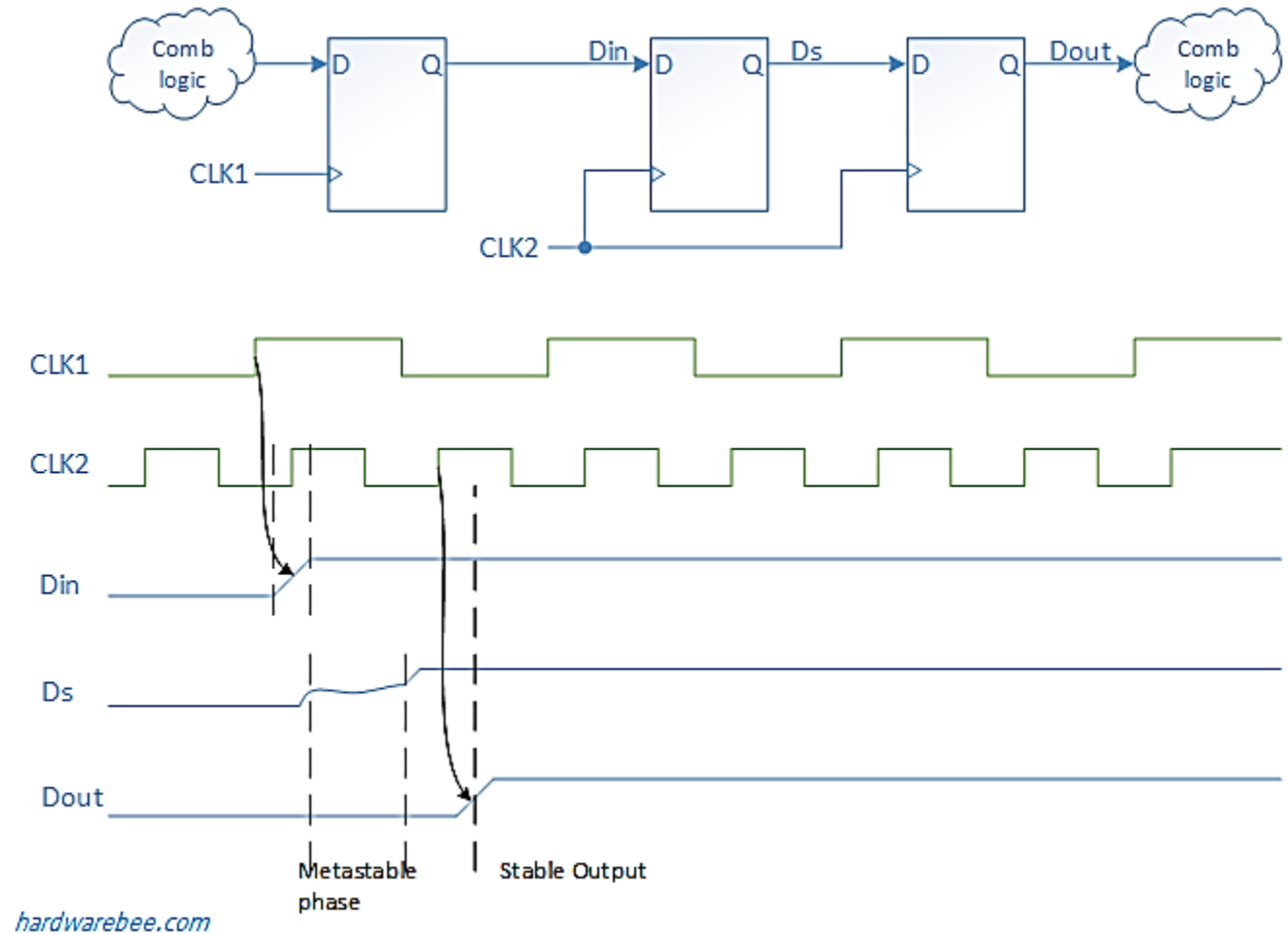
Setup and Hold time

- **Setup time** is the minimum amount of time *before* the clock event by which the input data must be stable for it to be latched correctly.
- **Hold time** is the minimum amount of time *after* the clock event by which the input data must be stable for it to be latched correctly.
- Any violation in this required time causes incorrect data to be captured and is known as a setup violation.



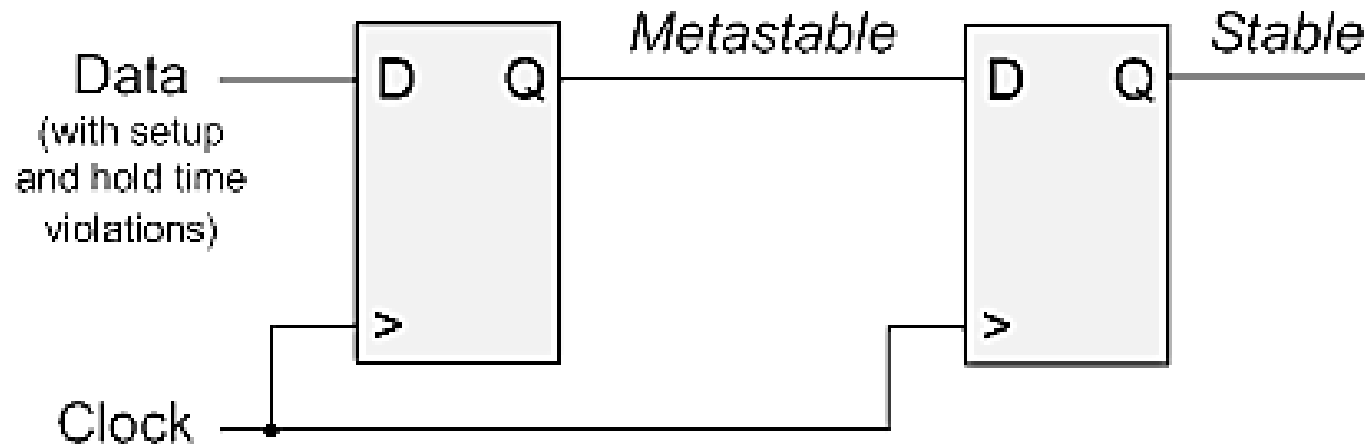
Metastable

- A metastable condition occurs when setup or hold times are violated.
- When happens, the digital electronics systems stuck for a period of time.



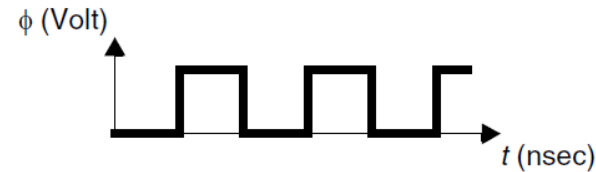
Avoiding Metastability

- It's advised to utilize one master clock signal or several clocks derived from this master clock.
- Inputs must be synchronized with the system clock before being applied to a synchronous system.

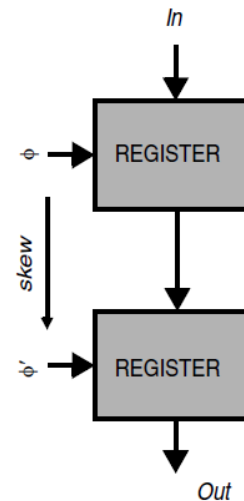


Clock distribution

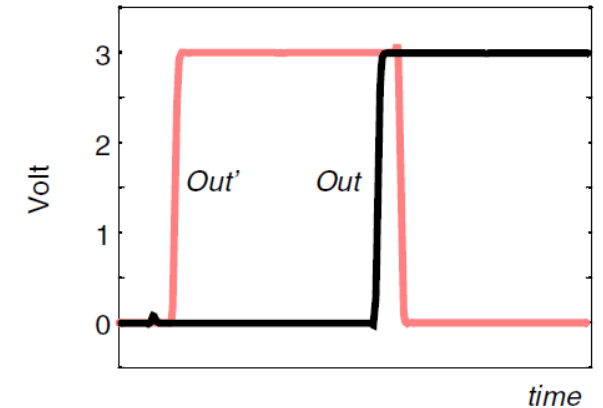
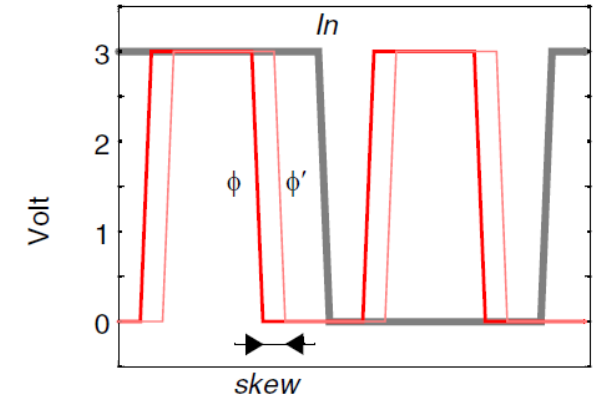
- Due to delays associated with routing the clock wires, it may happen that the clocks become misaligned with respect to each other; clock delay or skew



(a) Ideal clock waveform



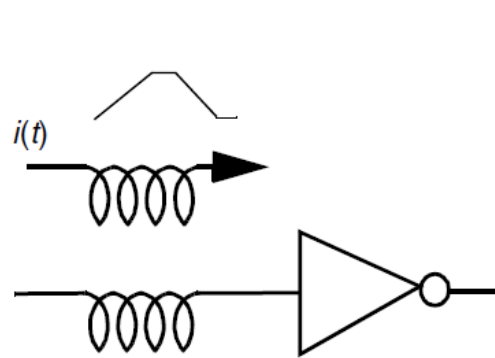
(b) Two cascaded registers



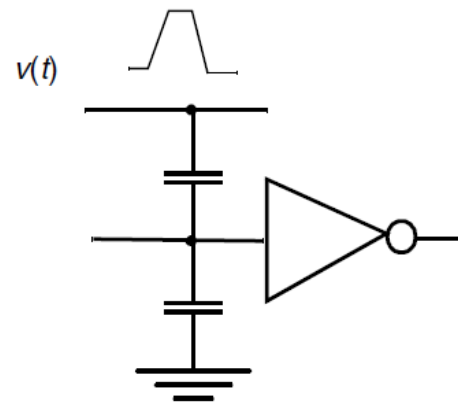
(c) Simulated waveforms

Functionality and Robustness

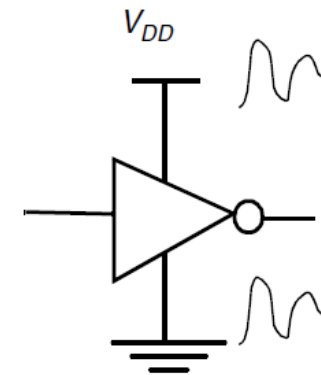
- Noise in digital circuits means “*unwanted variations of voltages and currents at the logic nodes.*”
- *Noise sources in digital circuits:*



(a) Inductive coupling



(b) Capacitive coupling

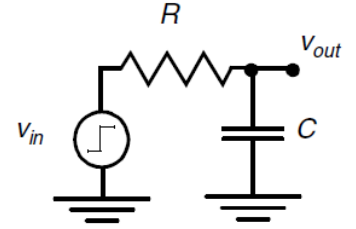


(c) Power and ground noise

Power and Energy Consumption

- It determines how much energy is consumed per operation and heat the circuit dissipates.
- Design decisions depends on:
 - power-supply capacity
 - battery lifetime
 - supply-line
 - sizing
 - cooling

Power and Energy Consumption



- Dissipation measures:

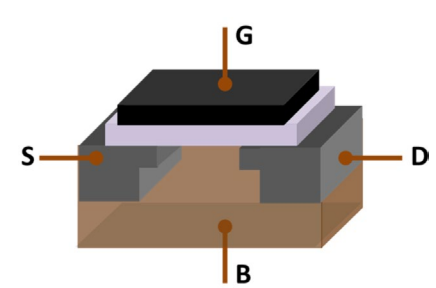
$$P_{peak} = i_{peak} V_{supply} = \max[p(t)]$$

$$P_{av} = \frac{1}{T} \int_0^T p(t) dt = \frac{V_{supply}}{T} \int_0^T i_{supply}(t) dt$$

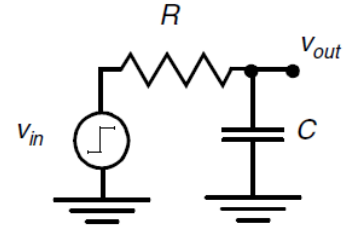
Where:

- $p(t)$ is the instantaneous power
- i_{supply} is current being drawn from the supply voltage V_{supply}
- *higher the number of switching events, the higher the dynamic power consumption.*

PDP (power-delay product): energy consumed by the gate per switching event



Energy Dissipation of First-Order RC Network



- Total energy delivered by the source

$$E_{in} = \int_0^{\infty} i_{in}(t) v_{in}(t) dt = V \int_0^{\infty} C \frac{dv_{out}}{dt} dt = (CV) \int_0^V dv_{out} = CV^2$$

- Energy stored in the capacitor

$$E_C = \int_0^{\infty} i_C(t) v_{out}(t) dt = \int_0^{\infty} C \frac{dv_{out}}{dt} v_{out} dt = C \int_0^V v_{out} dv_{out} = \frac{CV^2}{2}$$

- Where is the other half?